Biogeographic Regionalization: Freshwater

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Objectives 1
Introduction 1
Previous Attempts to Define Freshwater Zones 2
Global Freshwater Fish Biogeographical Regions 5
Nearctic 5
Neotropical 5
Palearctic 6
Ethiopian 7
Madagascar 7
Sino-Oriental Region 7
Australian 8
Conclusion 8
References 9

Abstract

Defining the number and geographical borders of regions containing similar organisms and high levels of endemism can shed light on the evolution and distribution of life on Earth. We provide an historical overview of studies delineating the global biogeographical regions of freshwater organisms, mainly focusing on fish, to understand whether aquatic and terrestrial organisms share similar distribution patterns. Then, we provide a geographical and biological description giving special attention to major biogeographical fish patterns and taxa present in each of the considered regions.

Glossary

Biogeographical region or zone a geographical area having similar biota.
Bioregionalization the process of delineating biogeographical zones, the way of defying its number and its geographical extent (or limits).
Endemic species a species (or a group of species from a given taxa or clade) that occurs in a single region (or geographical area) and nowhere else on Earth.
Endorheic river basin a river basin having no outlet to the sea.
Freshwater organisms organisms living and dispersing in fresh waters during part or their whole life-cycle.
Transition areas areas having biotas shared between two or more biogeographical regions due to diffuse boundaries and an interchange of species; these areas may be attributed to one or another biogeographical region, or even appear as individual biogeographical regions due to the peculiarity of its biota, depending on the group of organism being studied and/or the methods of bioregionalization.
Taxon or taxa an organism (taxon) or a group of organisms (taxa) sharing similar characteristics, pertaining to a given taxonomic group (e.g., same genus or family) or evolutionary branch.

Objectives

- Summarize previous and temporal knowledge on the global scale of freshwater bioregionalization.
- Describe major biogeographical zones, giving details on the diversity of freshwater fish fauna.

Introduction

Naturalists have long documented major differences in fauna and flora of different landmasses around the world (reviewed by Cox (2001)). Their efforts in the Nineteenth century were dedicated to identify biogeographical regions and define their geographical

limits. Defining the geographical range, the positioning and the number of biogeographical regions (biogeographical regionalization, hereafter) is of primarily importance because it is strictly linked to the way life is distributed, the underlying causes of the origins of species on Earth and its geographical spread over time (Brown and Lomolino, 1998). The bioregionalization process was hence extensively produced during the 18th and 19th centuries because early naturalists were struggling to understand and organize the distribution of organisms on Earth surface.

The first world-wide system of biogeographic regions was elaborated by Augustin de Candolle, a Swiss botanist (1778–1841) who separated the world in plant "formations" (i.e., biomes) (Cox, 2001). He provided 20 continental and island regions having distinct plant species and endemism levels due to differences in climate conditions (Cox, 2001). Despite his pioneering effort, de Candolle did not provide any maps to visually express his findings. Philip Lutley Sclater (1829–1913), a British zoologist interested in passerine birds and a good friend of Charles Darwin, proposed in 1857 a biogeographical framework reflecting the origin and the development of biotas, hence "the most natural primary ontological divisions of the earth's surface" (Brown and Lomolino, 1998). He divided the world into six major biogeographical regions assuming that species must have been created in the regions where they were present, and hence that there were multiple origination centers of biodiversity (Brown and Lomolino, 1998).

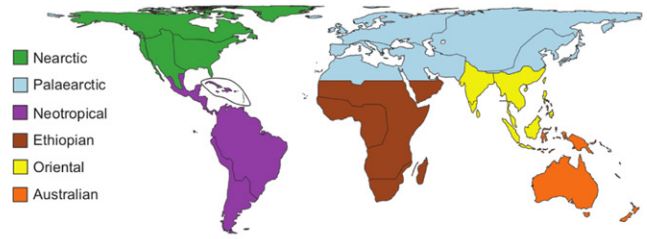
Alfred Russel Wallace (1823–1913), the father of zoogeography and one of the greatest British naturalists, built upon and greatly expanded the pioneering Sclater's scheme. Based on all terrestrial vertebrates (but excluding flying mammals), he divided the world into six regions, each further divided into a few subregions, and provided a very detailed map showing the limits of its biogeographical regions (Brown and Lomolino, 1998; Whittaker *et al.*, 2013). According to him, the biogeographical regions were Nearctic, Neotropical, Palaeartic, Ethiopian, Oriental, and Australasian (**Fig. 1**). This framework was a cornerstone in biogeography and a widely adopted model for biogeographical regions. The growing volume of biological studies and data in the subsequent decades enabled a series of refinements in the model proposed by Wallace. Those new updates refer overall to the number of biogeographical regions (e.g., the proposition of new ones and the splitting of others), to the geographical extent and the limits of the transition area between regions, to their nomenclature, and to the inclusion of other groups of organisms (Cox, 2001; Holt *et al.*, 2013; Kreft and Jetz, 2010; Morrone, 2015, 2002; Procheș and Ramdhani, 2012; Ribeiro *et al.*, 2014). Indeed, the differential levels of dispersal constraints, the evolutionary history of distinct taxa and the geological history of their habitats may limit the widespread adoption of Wallace's biogeographical regions for distinct groups of organisms.

Strictly freshwater organisms, like fish, are ideal models for biogeographic investigation because their dispersal depends on changes in the river network, which are driven by geomorphological processes and climatic events. Strictly freshwater faunas are geographically isolated in river basins because terrestrial landmasses, oceans, mountain ranges and basin divides are strong barriers to dispersal (Hugueny *et al.*, 2010; Rahel, 2007), hence their evolutionary history are linked to the historical changes in the river network (Bertuzzo *et al.*, 2016; Burridge *et al.*, 2008; Grant *et al.*, 2007; Oberdorff *et al.*, 2019; Tonkin *et al.*, 2018) and therefore the biogeographic history of a drainage basin (Banarescu, 1992; Myers, 1938). As a result, freshwater organisms could not follow Wallace's six biogeographic regions based on terrestrial mammals. It is therefore important to understand the number and limits of biogeographical regions to shed light on how the processes of diversification and evolutionary history have resulted in unique freshwater faunas within these biogeographic regions. In this chapter, we reviewed the previous studies aiming to describe freshwater biogeographical regions, their geographical limits and major organisms.As freshwater fish are among the most important organisms for both ecosystems and human beings, which account for one-fourth of the world vertebrates, we mainly focus on this group of organisms, but we also provided some information on other groups with the major biogeographical patterns without going deep on the details. (Box 1).

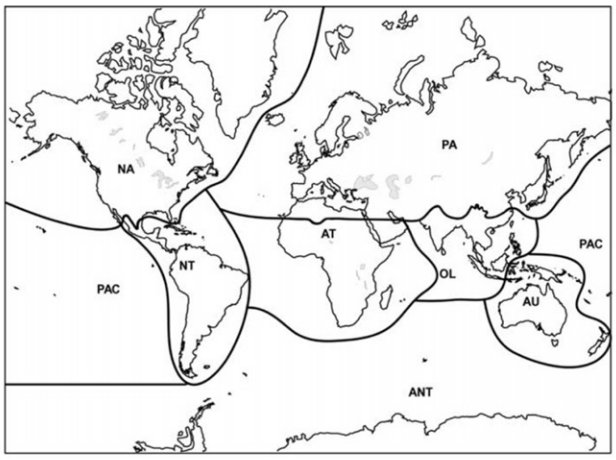
Previous Attempts to Define Freshwater Zones

Since the proposal of six major biogeographical zones by Sclater, and Wallace (hereafter SW zones), the limits of these geographical regions have remained relatively constant over time. Most authors adopted the proposed zones based on terrestrial vertebrates, and described the number of species, genus, families of freshwater organisms without adopting a quantitative and analytical framework to define the delineation of zones (Matthews, 1998). As a result, this is a major biogeographical model used until now for freshwater organisms (Balian *et al.*, 2008; Berra, 2007; Matthews, 1998). Examples of such widespread adoption are the works of Darlington (1957); Berra (2007), and the much more recent Freshwater Aquatic Diversity Assessment (FADA; Balian *et al.*, 2008).

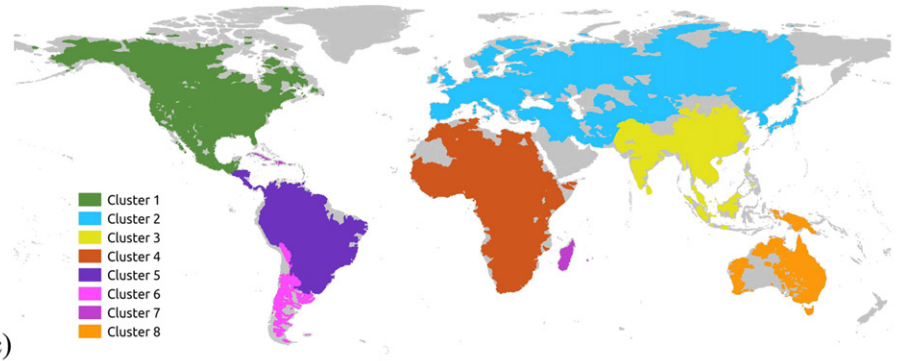
The FADA objective was to provide the most comprehensive evaluation of freshwater diversity of life, to consolidate the current knowledge on the distribution of living organisms, to estimate the global number of species, genus and families of aquatic organisms, to point geographical areas with high endemism levels and to provide an extensive data set (Balian *et al.*, 2008). Their assessment covered a large range of freshwater organisms, ranging from invertebrates (e.g., nematodes, bryozoans, and insects), vertebrates (e.g., fish, birds, and mammals) to macrophytes, thus, it represents an outstanding work to consolidate global freshwater diversity of life. The authors, however, acknowledged the difficulties in delineating global biogeographical regions and adopted the six SW regions (Cox, 2001; Wallace, 1876) plus two complementary zones, the Antarctic region (ANT) and Pacific Ocean region and its Oceanic Islands (PAC). Whereas ANT region encompasses the Antarctic continent, Antarctic and subantarctic islands south of the Antarctic convergence, the PAC region included islands in North and South Pacific Ocean, plus the Bismarck Archipelago, Vanuatu, the Solomon Islands and New Caledonia. They also raised some issues concerning transitional zones (between Palaeartic and Oriental regions, Oriental and Australasian, and Nearctic and Palaeartic regions), and referred to slightly distinct names from those proposed in SW regions for homogenizing purposes (Balian *et al.*, 2008) (**Fig. 1**).


Nearctic
Palaearctic
Neotropical
Ethiopian
Oriental
Australian

a)


NA
PA
AT
PAC
NT
OL
PAC
AU
ANT

b)


Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8

c)



d)

Box 1 Summary of the predominant methods applied in ecological and biogeographical literature to define the biogeographical zones.

The increased effort in delineating regions lead to confusion on the appropriate biogeographical model to be adopted, and mostly on the methods used to define those regions (Kreft and Jetz, 2010). Fortunately, we have witnessed in the last decades an increasing upsurge of biogeographical regionalization for particular taxa, based on molecular phylogenetic analyzes and advanced statistical methods (Costello *et al.*, 2017; Holt *et al.*, 2013; Kreft and Jetz, 2010; Vilhena and Antonelli, 2015). Biogeographic regions can be defined in various ways, ranging from using expert knowledge as often performed during the last century to more recent data-driven approaches (Edler *et al.*, 2017; Holt *et al.*, 2013; Kreft and Jetz, 2010; Vilhena and Antonelli, 2015). Nowadays, bioregionalization is based on species occurrence data and often rely on turnover among assemblages and distance-based clustering methods (e.g., González-Orozco *et al.*, 2013; Kreft and Jetz, 2010). More specifically, Simpson's dissimilarity index (βsim) has been widely used to quantify pairwise dissimilarity among grid cells (Kreft and Jetz, 2010; Leprieur and Oikonomou, 2014), mainly because it reflects pure species turnover and is robust to differences in species richness (Baselga, 2010), two desirable properties in bioregionalization studies (Kreft and Jetz, 2010). Recently, a network-based method based on a community detection algorithm known as Infomap has been put forward for identifying biogeographic regions (Edler *et al.*, 2017; Vilhena and Antonelli, 2015). In addition, Maestri and Duarte (2020) using a fuzzy logic-based approach introduced the concept of ecoregion where most of the resident species stem from one or a few in situ radiations, with the evolutionary transition zones showing high phylogenetic turnover, and therefore having a low affiliation to a single evoregion. Mapping those regions in the phylogenetic tree quickly reveals the timing and location of major shifts of biogeographic regions. Irrespective of the adopted methods, better delineations of regions and its drivers (Ficetola *et al.*, 2017) can greatly increase our understanding of the evolution and distribution of life on Earth.

Matthews (1998) was one of the first to confront the SW zones for freshwater organisms based on an "objective quantitative analysis of similarities and differences in fishes among smaller units of geographical space". In this framework, he a priori divided the whole world in 52 terrestrial regions mostly corresponding to large river basins and then used fish range maps to compose a worldwide presence/absence matrix of freshwater fish family per river basin. Both geographical regions and the range maps of fish families had been gathered from Berra (1981). By using a cluster analysis based on the Jaccard's similarity index, Matthews were able to reconstruct nine major clusters that overall matched the biogeographical regions defined by SW. Three major clusters coincided with Nearctic, Palaeartic, and Oriental regions from SW's model, the only exception being the far-north Africa (Sahara desert region) which was not included in the Palearctic region (**Fig. 1**). This was also the case for the Arabian Peninsula, but (Matthews, 1998) had no information of fish and rivers from this desertic region in his analysis. In contrast to SW delineation, the Africa, or Ethiopia region, derived from Matthews (1998) did not include Madagascar island which appeared in a separated cluster linked to the West Indies in Caribbean due to the shared Cyprinodontidae and Cichlidae families. Two distinct clusters composed the Latin America: the Neotropical region *per se* formed by Central America, northwest and east South America; and another region (cluster 6) of South America south and west of the Paraguay-Parana river basin because of its depauperate fish fauna. Finally, the Australia region was also composed of two distinct clusters, one based on Australia and islands southeast of Wallace's Line, and another separating New Zealand due to the absence of freshwater fish families. Overall, his impressive and beforehand, though classified by himself as "naive" (*sensu* Matthews, 1998), analysis captured almost the same zoogeographic regions defined by SW one hundred years ago.

Another important contribution to bioregiolalization studies on freshwater is the seminal paper of Abell *et al.* (2008). These authors argued that previous efforts to divide the world into major biogeographical zones ("in the level of continent", *sensu* Abell *et al.*, 2008) had limited values for conservation effort planning and performed a much more detailed delineation of freshwater fish ecoregions. The concept of ecoregion was depicted by them as a large area containing much more similar freshwater environmental conditions, freshwater species and dynamics, and having distinct natural freshwater communities and species compared to the surrounding areas (Abell *et al.*, 2008). In this way, they argued, their freshwater ecoregions provide information on the underlying ecological and evolutionary processes driving species composition patterns. The model organism used to compose these ecoregions were strictly freshwater fish for a few reasons. Overall, this was the best studied group in global freshwaters, the distribution information was the most complete when evaluating regions other than western Europe and North America, and because saltwater is an effective barrier of dispersion for strictly freshwater fish organisms, making fish better zoogeographic indicators than freshwater invertebrates (Abell *et al.*, 2008). Due to the challenges of leading such a comprehensive global assessment, the delineation of global fish ecoregions was far more heterogeneous than the previous global scale effort (Matthews, 1998). For instance, the data sets and the strategies used to delineate ecoregions varied between regions, ranging from

Fig. 1 Examples of biogeographical regions according to (a) Sclater-Wallace terrestrial regions and the adopted (b) Freshwater Aquatic Diversity Assessment (FADA) regions, which coincides with the terrestrial regions. (c) Global clusters defined by Matthews (1998) with data from freshwater fish families and adapted using global drainage basins (Tedesco *et al.* 2017) to improve clarity. (d) Ecoregions delineated by Abell *et al.* (2008) using fish data from multiple origins and expert-based knowledge. Reproduced from (a) Kreft, H., Jetz, W., 2010. A framework for delineating biogeographical regions based on species distributions. Journal of Biogeography 37, 2029–2053. Available at: https://doi.org/10.1111/j.1365-2699.2010.02375.x. (b) Balian E., Segers H., Lévèque C., Martens K., 2008. The freshwater animal diversity assessment: An overview of the results Hydrobiologia. 627–637. Available at: **XXXXXXXXXXX**. (c) Modified from Matthews, W.J., 1998. Patterns in freshwater fish ecolo, Springer Science & Business Media. (d) Abell, R., Thieme, M.L., Revenga, C., *et al.*, 2008. Freshwater ecoregions of the world: A new map of biogeographic units for freshwater biodiversity conservation. BioScience 58, 403–414. Available at: https://doi.org/10.1641/B580507.

presence/absence species list per sub-drainage basins in well-know regions (e.g., USA, and Mexico), species list from irregular sampled catchments (e.g., Congo river basin) to delineation by leading ichthyologists when there were low available information on fish (e.g., Africa, sub-provinces of New Guinea, Southeast Asia and southern Europe).

By using more than 13,400 freshwater fish species of which 6,900 were endemic, Abell *et al.* (2008) defined 426 distinct ecoregions covering the whole world. They identified high levels of fish endemism and richness in large portions of the Congo river basin, the southern Gulf of Guinea drainages and the lakes Malawi, Tanganyika, and Victoria in Afrotropical region, the Pearl (Zhu Jiang) river and neighboring systems, Brahmaputra, Ganges, Mekong (plus surrounding drainages) and Yangtze river basins in Oriental Region, and in large portion of Neotropical Region (Orinoco, Amazonia, Tocantins-Araguaia, São Francisco, and Paraná river basins). Despite the limitations imposed by challenges listed above, this impressive work remains as an important and precise delineation of freshwater ecozones at a more regional spatial scale.

A recent study at the global scale on freshwater fish bioregionalization has identified the major biogeographical regions based on a network framework (Leroy *et al.*, 2019). In bioregionalization studies, a network is composed of species and sites as nodes and the links between nodes are represented by the occurrence of species in each given site (Vilhena and Antonelli, 2015). Leroy *et al.* (2019) applied this rationale on the most comprehensive global database of fish occurrence (Tedesco *et al.*, 2017), using occurrence information on 11,295 strictly freshwater species in 2,581 drainage basins distributed all over the world. They used this dataset to apply a hierarchical community-detection algorithm (Rosvall and Bergstrom, 2008) and aggregate the network nodes (i.e., river basins and species) into distinct clusters (i.e., biogeographical regions), which could be further divided (i.e., subregions) in a nested hierarchy (Rosvall and Bergstrom, 2008; Vilhena and Antonelli, 2015). Besides defining the number and the major limits of the biogeographical regions by the aggregation of river basins, these authors also calculated the participation coefficient for each node, indicating the connection degree of nodes to different zones, hence clearly stating the transition zones and shared species between regions (Bloomfield *et al.*, 2018; Leroy *et al.*, 2019).

Overall, they found two supercontinental regions (New World and Old World) with almost half of the total analyzed world freshwater fish species, of which nearly 100% of species, 99% of genera, and 80% of families were endemic. These supercontinental regions were further separated into Nearctic and Neotropical (New World), and Paleartic, Ethiopian, Sino-Oriental, Australian and Madagascan (Old World) major biogeographical regions (**Fig. 2**). As the work of Leroy *et al.* (2019) represents the most up-to-date analytic bioregionalization for an aquatic organism as far as we know, we used their model to describe and characterize each of the biogeographical regions with a particular look for strictly freshwater fish species.

Global Freshwater Fish Biogeographical Regions

Nearctic

This region encompasses Northern Mexico and North America up to Alaska (Leroy *et al.*, 2019). The fish fauna is composed of more than 842 strict freshwater species (Leroy *et al.*, 2019), reaching more than a thousand species when including saltwater tolerant and migratory species (Lundberg *et al.*, 2000). Cyprinidae, Percidae, Poecilidae and Catostomidae are dominant families, representing more than 50% of fish species, but other predominantly marine families occur (Cottidae, Atherinidae, Clupeidae and Moronidae) and represent an important component of fish fauna (Lévêque *et al.*, 2008; Lundberg *et al.*, 2000).

The largest river basins are Mississippi, Mackenzie, St. Lawrence, Nelson, Yukon, Columbia, Colorado and Rio Grande, but the largest diversity of freshwater organisms (amphibians, birds, crayfish and mammals other than fish) is concentrated in the Southern portion (Gulf of Mexico and Atlantic drainages) of this region (Griffiths, 2010; Lundberg *et al.*, 2000; Tisseuil *et al.*, 2013). The Mississippi river basin contains more than 375 fish species, among which 130 endemics (Lévêque *et al.*, 2008). This is mostly because geology and climate have been relatively stable during the Cenozoic and due to differential rates of northward colonization in post-glacial periods for freshwater and terrestrial organisms (Dobrovolski *et al.*, 2012; Griffiths, 2015; Lundberg *et al.*, 2000). Although low total diversity, western (e.g., Colorado and the Great Basin drainages) and small Mexican basins present high fish endemism levels (Tisseuil *et al.*, 2013) due to high isolation, mountainous and arid conditions (Lundberg *et al.*, 2000).

Neotropical

It comprises basins from the whole South America and Mesoamerica up to Southern Mexico. The largest river drainage basins are the Amazon, Orinoco, Tocantins, La Plata (Paraná-Paraguay system) and São Francisco rivers. The transition zone between Neotropical and Nearctic regions lies at Guerrero, Oaxaca, Puebla and Veracruz Mexican states, where a few small bodied-sized fish species from Nearctic and Neotropical regions incurred in the opposed regions (Leroy *et al.*, 2019). The major fish families involved in this transition zone are Poeciliidae and Characidae and most species are of restricted distributional ranges near the regional boundaries.

The Neotropical region contains 85% of the New World supercontinental region and 42% of worldwide strictly freshwater fish species, totaling 4,761 fish species with 99.5% (4,736) of endemism (Leroy *et al.*, 2019). Recent estimates suggest this number will further increase as about two new species are described per week, reaching total estimates of $>9{,}000$ species (including freshwater and marine) in the whole region (Albert *et al.*, 2020; Reis *et al.*, 2016). The most important lineages are Characiformes, Siluriformes, Cichliformes, Cyprinodontiformes, and Gymnotiformes, but Neotropics also includes other ancient lineages such as Osteoglossiformes and Dipnoi and many other groups derived from marine ancestors (e.g., Belonidae, Tetraodontidae and Potamotrygonidae) (Albert *et al.*, 2020; Dagosta and Pinna, 2018; Lévêque *et al.*, 2008). The largest number of freshwater fish families is found in

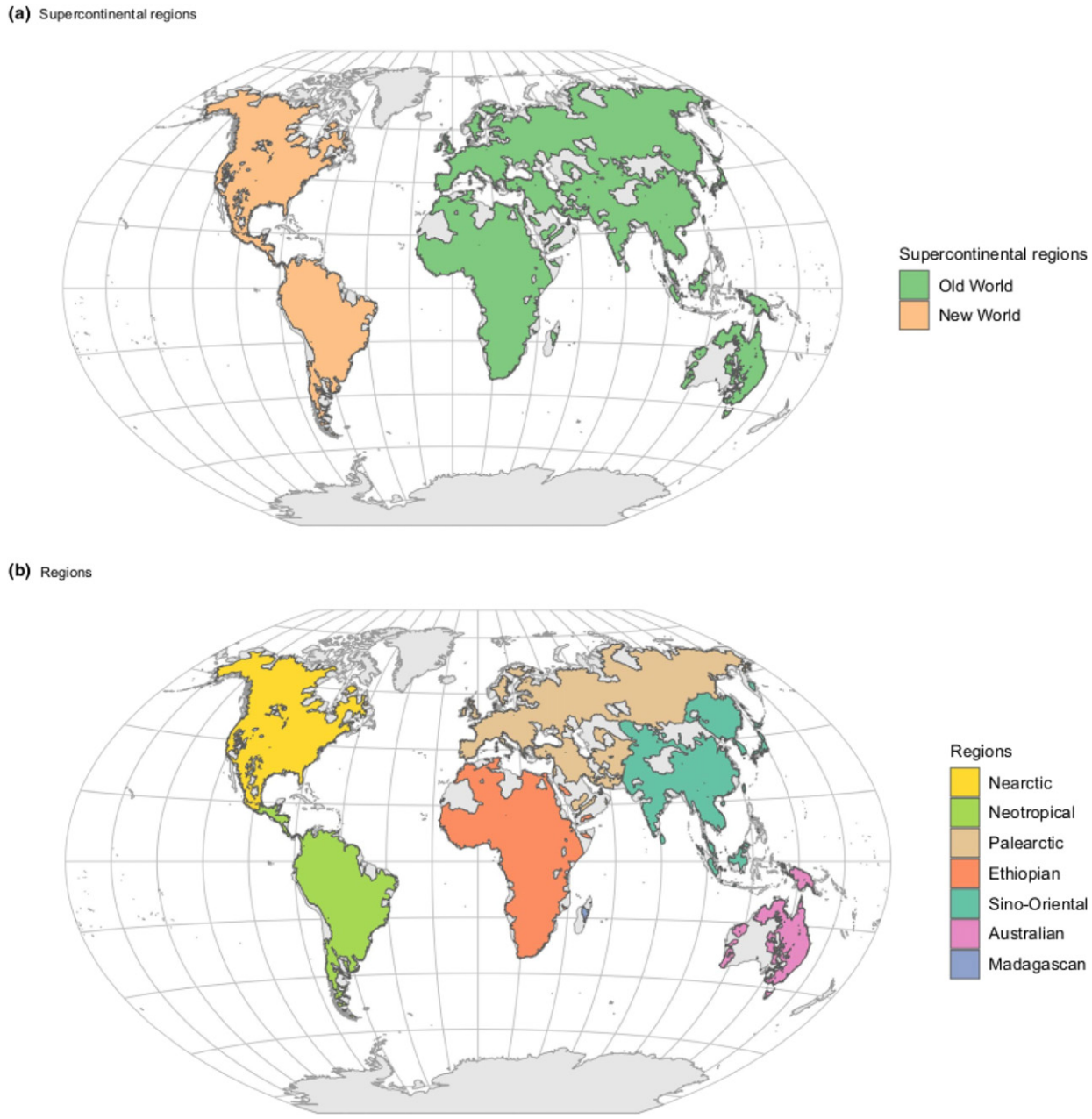


Fig. 2 Major biogeographical realms and regions defined for strictly freshwater fish based on a network algorithm. Reproduced from Leroy, B., Dias, M.S., Giraud, E., *et al.*, 2019. Global biogeographical regions of freshwater fish species. Journal of Biogeography 46, 2407–2419. Available at: https://doi.org/10.1111/jbi.13674.

Neotropics (43) (Lévêque *et al.*, 2008), almost, and characoids and siluroids are by far the most diverse group of organisms. The major center of endemism is located in the tropical region of South America, mainly at the Amazon, Orinoco river drainage basins and those coastal basins in the Guiana Shield (Albert *et al.*, 2020; Albert and Reis, 2011). The Amazon river drainage basin alone contains more than 2,400 fish species in its high diverse habitats (Albert *et al.*, 2020; Dagosta and Pinna, 2017, 2019; Jézéquel *et al.*, 2020; Oberdorff *et al.*, 2019), most of which are endemics, and including surrounding basins increases estimates to incredible 4,214 fish species (Dagosta and Pinna, 2019).

Palearctic

The Palearctic region is the second largest region on Earth and covers the entire area of Europe, Central Asia (up to Pakistan and Kazakhstan) and Siberia. Ten major river systems drain the region; major lakes include the Black and Caspian Seas and Lake Baikal,

the world's oldest and deepest lake. The transition zone between the Palearctic and Sino-Oriental regions lies beyond the Himalayan Mountains and the Gobi desert (Leroy *et al.*, 2019) (see also below).

Twenty-seven families and about 492 species of freshwater fishes occur in the region (4.4% of the world), dominated by the families of Cyprinidae, Salmonidae, Balitoridae, Cobitidae, Gobiidae, Bagridae, Sisoridae, Abyssocottidae, Cichlidae and Cyprinodontidae. Palearctic shows the lowest degree of freshwater fish endemism (88%) among the other biogeographical regions. The Palearctic appears to be the most speciose for most freshwater animal species, except for vertebrates (Balian *et al.*, 2008), but the reason for this fact is probably the intense taxonomic expertize and research efforts centered in this region, with the highest diversity of freshwater insects being recorded from the Palearctic (20%).

There is a clear trend of increasing richness and endemism from high latitudes in the Arctic region toward the south of this land mass with the European lotic animal diversity richness showing a strong decline with increasing latitude (Dehling *et al.*, 2010; Tisseuil *et al.*, 2013). Biogeographic evidence indicates that the northern areas had been glaciated many times during the Pleistocene, the northern part being covered either by ice or by permafrost, while the southern part (mainly areas surrounding the Mediterranean Sea, i.e., Peri-Mediterranean areas) was spared (Hewitt, 2000; Reyjol *et al.*, 2007). Freshwater fish, and potentially many other freshwater organisms, recolonized northern areas during the interglacial and/or post-glacial periods, mainly from refugia in the Danube river and other unglaciated, southern peninsula areas (Iberian, Italian and Balkan) in western Paleartic. All these refuge area have high richness and endemism level for fish (Griffiths, 2006; Oikonomou *et al.*, 2014; Reyjol *et al.*, 2007) and other freshwater organisms, but with differential extent, lending the second highest importance to historical/dispersion factors in explaining these diversity gradients (Tisseuil *et al.*, 2013).

Ethiopian

The entire African continent, including those north of the Sahara, and a few basins in the Arabic Peninsula, comprises the Ethiopian region, the second most diverse region. Due to the high climatic and hydrological variability, a great variety of freshwater habitats compose this region (e.g., temperate and tropical rivers, wetlands, floodplain rivers and marshes in equatorial climates, shallow and rift lakes, and endorheic river basins) (Fouchy *et al.*, 2019; Lemoalle, 2017). Major drainage basins are the Congo River in the tropical western Africa, where the highest diversity of fish is found (Abell *et al.*, 2008; Leroy *et al.*, 2019), the Nile, Niger, and Tchad (endorheic) rivers in northern, and the Zambezi and Karoo-Orange rivers in southern Africa, where subtropical and semiarid climate occur (Fouchy *et al.*, 2019). Eastern Africa is composed of tropical and subtropical rivers, endorheic basins, and large lake systems over the East African Rift, a remarkable hotspot of biodiversity (Leroy *et al.*, 2019).

This region has more than 2,683 strictly freshwater fish species distributed in about 331 genus and 34 families, of which ~95% and 50% are endemic (Leroy *et al.*, 2019). Overall, 23–24% of all global fish diversity is present in the Ethiopian region (Leroy *et al.*, 2019; Lévêque *et al.*, 2008), reaching more than 3,360 fish species including brackish organisms (Lévêque *et al.*, 2008). The families Cypridinae, Alestidae, Mormyridae and a few families from Siluriformes and Cyprinodontiformes are the most common taxonomic groups in the whole continent; Cichlidae is the richest family, but most of these species are restricted to the East African lakes (Lévêque and Paugy, 2017). The Ethiopian fish assemblages include groups restricted to fresh waters (e.g., Mormyrids; about 55% of recorded species), groups confined to freshwaters but salt-tolerant (i.e., with occasional capabilities to cross narrow sea water barriers; e.g., some species from Cichlidae and Cyprinodontiformes; 32% of recorded species), and groups from extant marine lineages that colonized inland waters (e.g., Latidae, Perciformes, Pleuronectiformes and Tetraodontiformes; 12% of recorded species) (Lévêque and Paugy, 2017). As the African continent has remained above sea level for more than 600 million years ago (since the Precambrian), a large proportion of freshwater fish families is endemic and represents ancient lineages (e.g., Protopteridae and Arapaimidae), some of them shared with Australian, Sino-Oriental and Neotropical regions. Whereas Cyprinidae and Siluriformes are shared between Ethiopian and Sino-Oriental, Siluriformes are abundant and native Cyprinidae are absent in Neotropics (Lévêque and Paugy, 2017).

Madagascar

Madagascar is the world's fourth largest island, being separated from Africa about 165 million years ago, about the same time as the breakup of Gondwanaland, and has been isolated from all other landmasses (most recently, India) for approximately 88 million years (Rabinowitz *et al.*, 1983). The island's large size, its varied topography combining with large differences in the annual amount and seasonality of precipitation on different island regions result in a substantial diversity of rivers, streams, and other freshwater habitats and high levels of endemism for many groups of organisms (Benstead *et al.*, 2003). Thus, Madagascar is recognized as a global hotspot for freshwater biodiversity (World Conservation Monitoring Center, 1998). Regarding strictly freshwater fish, two families (Bedotiidae and Anchariidae), 9 genera and 18 species are endemic to the Madagascar island, mainly found in eastern forested regions (Leroy *et al.*, 2019), and about one quarter of endemic species are only known from their description type locality (World Conservation Monitoring Center, 1998).

Sino-Oriental Region

The Sino-Oriental region is the third richest region in the world, which includes south-eastern Asia from India to Borneo, most of China and Mongolia, Korea and Japan. There are 2,495 strictly freshwater fish in the Sino-Oriental region, which account for 22%

of the world freshwater fish species (Leroy *et al.*, 2019). The fish fauna belongs to ca. 50 families of primary and secondary freshwater fishes, of which, the most diverse are the minnows (Cyprinidae), loaches (Cobitoidea), and families from catfishes (Siluriformes). However, the number may be underestimated due to the continuous discovery of new fish species every year. The Sino-Oriental region is also one of the regions with the highest degree of endemism (98%, Leroy *et al.*, 2019). Compared to other taxa, the Sino-Oriental region has a distinctive freshwater fish fauna throughout its geographical extents. From the Stanovoï mountains in northern China to the northern-west boundary beyond the Gobi Desert and the Himalayas, the Sino-Oriental region shares many families with the Palearctic (e.g., Cyprinidae, Nemacheilidae) and the Ethiopian (e.g., Cyprinidae, Cyprinidae) regions, but shares only few families with the Australian region in its southeast borders (e.g., Gobiidae, Helfman, 2013). Other groups (e.g., mammals, birds) are limited to a boundary extending longitudinally from the Himalayas to the Chinese Sea (Leroy *et al.*, 2019). The peculiarity of the Sino-Oriental region in terms of fish was caused by two main factors. First, the entire region was prone to fish speciation since the Eocene (55 Ma) due to the high diversity of aquatic habitats combined with the repeated rearrangement of rivers through glaciers freezing and melting (Kang *et al.*, 2014; Leroy *et al.*, 2019). Second, different from other groups for which tectonics and climate may represent soft barriers to dispersal, the mountain barriers were much more stringent in defining boundaries for freshwater fishes. Thus, the fish transition zone between Sino-Oriental and Palearctic is not located near areas of recent tectonic merging as reported for other terrestrial groups (Leroy *et al.*, 2019). In addition, this region is composed of two major sub-regions, (i.e., Oriental and Sinean), which were frequently identified as distinct regions (Leroy et al., 2019).

The Oriental sub-region is limited in its western border by the Ghaggar (India) and Indus (Pakistan) rivers, by the Himalayas in the north, in the east by the Indo-China Peninsula and includes rivers from Java, Borneo and Sumatra islands. The Indian subcontinent, due to its diverse natural water resources, is a diverse region concerning freshwater organisms, with 765 freshwater fish species reported (Sarkar *et al.*, 2012, 2008). The Ganga River is the largest and also most important inland river in the Indian subcontinent, containing more than 178 freshwater fish species distributed in 37 families (dominated by Cyprinidae) and 89 genera (Ganga River Basin Environment Management Plan, 2012). Another important river of this sub-region is the Lancang-Mekong river. Lancang (upper part) and Mekong (lower part) are different names referring to the same river (hereafter, Lancang-Mekong River). It's one of the most important rivers flowing through China and the Indo-China Peninsula (through Myanmar, Thailand, Laos, Cambodia, and Vietnam before flowing into the South China Sea; (Liu *et al.*, 2007)). The Lancang-Mekong River and its tributaries constitute one of the most important freshwater hotspots in the world (Soukhaphon *et al.*, 2021): with its more than 1000 fish species (Coates *et al.*, 2003) and the many newly described ones in the last decade, it has been reported as one of the richest river (Baran *et al.*, 2012; Dudgeon, 2000) on Earth after the Amazon and the Congo rivers.

Although the Sinean sub-region was identified more as a transition zone for other groups (Morrone, 2015; Proches and Ramdhani, 2012), it hosts a rich freshwater fish fauna (963 species) with 72% of endemism level (Leroy *et al.*, 2019). China, as one of the megadiverse countries with a wide variety of freshwater fish, constitutes most part of this sub-region. The current distribution of China's freshwater fish fauna were formed in the late Tertiary to early Quaternary during the uplift of the Qinghai-Tibetan Plateau and the global climate cooling. During periods of glaciations, fish species retreated southwards, which favored frequent land connections between the south-eastern mainland of China and western Pacific islands (Kang *et al.*, 2014). Kang *et al.* (2014) reported a total of 920 fish species (302 genera, 54 families and 21 orders), of which 675 (73%) belong to Cypriniformes (the most dominant group in China). More recent estimates indicate 1,651 freshwater fish species in total, 63% (1,034) of endemism (He *et al.*, 2020). Another important part of the Sinean sub-region is the islands of Japan. A total of 312 fish species or subspecies have been recorded in Japanese inland waters, including 50 exotic species and 90 strict freshwater fishes (Katano and Matsuzaki, 2012). Gobiidae and Cyprinidae are the two main families, which constitute about 40% and 25% of Japanese freshwater fishes, respectively. Fish diversity increased gradually from the eastern regions to the western regions of Japan (Watanabe, 2012).

Australian

As clustered by Leroy *et al.* (2019), the Australian region covers Australia, Tasmania and Papua-New Guinea. The boundary between Australian and Sino-Oriental includes the entire Wallacea region: the Sino-Oriental extends to the limits of the Sunda shelf, and the Australian extends to the limits of the Sahul shelf (Leroy *et al.*, 2019). There are a few major rivers flowing through this region, including the Fly River in New Guinea, and the Murray-Darling river in Australia. The Australian region is relatively depauperate but highly unique compared with other fish regions, largely due to Australia's dry and arid climate and long isolation from other regions (Shelley *et al.*, 2019). According to Leroy *et al.* (2019), there were only 90 strictly freshwater fish in this region (about 0.8% of the world's freshwater fish fauna), but most of them (87 species or nearly 97%) are endemic.

Conclusion

The reviewed studies show the S-W terrestrial framework share many similarities with the major freshwater biogeographical region, including the major six regions and the contrasting limits being at the Ethiopian-Palaeartic, Palaeartic-Sino-Oriental transitional zones. Still, even though numerous sub-regions and ecoregions can be identified in a fine scale geographical level, biogeographical zones of fish are consistent even using coarse taxonomic (i.e., families), a limited number of spatial units (i.e., only major drainage basins) and distinct clustering algorithm (Matthews, 1998).

Numerous topics of further development might be developed in the future. Although the global bioregionalization based on taxonomic data (Leroy *et al.*, 2019) is so far the most comprehensive assessment for freshwater organisms, future analytical improvements can be provided and data included to evaluate whether these results are consistent with other freshwater organisms (Holt *et al.*, 2013). Determining the drivers of biogeographical boundaries can also shed light on the processes related to the evolution and dispersal of freshwater organisms over time (Daru *et al.*, 2017; Ficetola *et al.*, 2017). Yet, prior to human intervention, no species of freshwater fish naturally occurred in all these biogeographic regions and only few species occurred in more than one region (Leroy *et al.*, 2019; Rahel, 2007). The delineation of biogeographic regions by freshwater organisms is therefore also important for identifying changes on the regions, especially under the increasing rate of freshwater species introductions and extirpation. Today, for instance, rainbow trout, brown trout, brook trout, mosquitofish, guppy, common carp and goldfish occur in all of the Earth's biogeographic regions (Lever, 1996) and most developed regions experience extirpation of species from its regional assemblages (Dias *et al.*, 2017), leading to homogenization of regional regions (Capinha *et al.*, 2015; Carmona *et al.*, 2021; Toussaint *et al.*, 2021). The bioregionalization studies in freshwater environments are certainly a promising avenue of research.

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